

StainlessCarbon Research Appendix

The evidence base behind the carbon & energy calculator and the two-slide executive summary: emission factors, methodology, regulation, company disclosures, competitor analysis, and the locked canonical model that every submitted figure traces to.

Live calculator: ritehrks.github.io/jindal-steel-case-study/website

Repository (19 research files, full sources): github.com/ritehrks/jindal-steel-case-study

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1. The canonical model — locked figures and rulings

Full text: research/22-canonical-model-v1.md · every figure in the deck and calculator traces here

During pre-submission audit we found three conflicts across the research base and resolved each with a written ruling. The rulings, not the underlying files, govern all submitted numbers.

- **CBAM dataset.** Two candidate datasets existed. We locked the traceable one: India stainless default **6.49 tCO₂e/t** base rising to **8.44** with the +10/20/30% no-data markup (Implementing Regulation (EU) 2025/2621, corrected 2026/1740); ETS price path €80→€109 (2026–30); phase-in 2.5%→48.5% (Directive (EU) 2023/959). Liability quoted on the analyst-standard pro-rata basis.
- **Jindal's CBAM-relevant intensity.** A single-plant Scope 1+2 figure (0.65) was rejected: CBAM counts company-wide direct emissions. Derived company-wide Scope 1 from BRSR absolutes: FY24 **1.70** (matches CDP exactly), FY25 1.53, FY26 ≈ **1.52 tCO₂e/tcs**. Because captive-power emissions count as indirect under CBAM Annex II, the realistic CBAM-direct basis is **0.9–1.1**; we quote 1.4–1.5 as the conservative bound.
- **Benchmark comparisons.** Jindal's S1+2 (1.76) must not be compared with Outokumpu's 1.8 (a cradle-to-gate figure including upstream Scope 3). Correct comparisons: cradle-to-gate 3.03 vs worldstainless global average ≈3.70; S1+2 1.76 vs Indian steel-sector ≈2.5.
- **Boundary rules.** R1: captive ferrochrome and captive power live inside Scope 1+2 — never double-counted as Scope 3. R2: scrap enters burden-free (ISSF recycled-content); end-of-life credit is a second output. R3: intensities per tonne of *crude* steel (derived: FY24 1.758 Mt, FY25 1.956 Mt, FY26 2.015 Mt); sales tonnes are never the denominator.
- **Renewable share.** 47% locked for FY26 full-year (assured ESG Factsheet basis); the "56% by Q3 FY26" figure is quoted only as dated momentum.

2. Emissions model specification and calibration

CO₂/t = S1[0.12 AOD process + 1.244·(1 - 0.4·cleanfuel) fuels & process heat + 1.10·(1-RE)·0.5·1.045 captive power]
+ S2[1.10·(1-RE)·0.5·0.710 grid electricity]
+ S3[(1 - scrap%) · ∑ m_i·EF_i] subject to scrap ≤ 80%, Cu/Sn bounds

Grade-wise virgin-charge chemistry (kg per tonne crude steel)

Grade	Cr %	Ni %	Mo %	HC FeCr (60% Cr)	Contained Ni	FeMo (70% Mo)
304 austenitic	18	8	–	300	80	–
316 Mo-grade	17	10	2	283	100	29
430 ferritic	17	–	–	283	–	–
2205 duplex	22	5	3	367	50	43

Triple validation. (1) *Audited anchor:* FY26 inputs (70% scrap, 47% RE, 30% clean fuel, mixed Ni) return S1+2 1.73 and total 2.99 vs the audited 1.76 / 3.03 — within ~2%; FY24 inputs return 4.03 vs disclosed 4.05. (2) *Industry curve:* outputs track the worldstainless scrap curve (85%→1.95, 50%→3.70, 30%→6.80). (3) *Grade physics:* nickel-free 430 lands near half the 304 footprint.

3. Emission factors: routes, alloys, India fuels and power

Files 01, 04, 05 · sources: worldsteel, Columbia CGEP, ICDA, Metso Outotec, ISSF, CEA v21.0, BEE PAT, IPCC 2006, MoEFCC BUR-3

Steelmaking routes (cradle-to-gate unless noted)

Route	tCO ₂ /t crude steel
BF-BOF (virgin)	~2.2 (1.8–3.0)
Scrap-EAF	~0.67
DRI-EAF (natural gas)	0.7–1.2
DRI-EAF (coal)	2–3
EU CBAM plant-gate benchmarks (reference)	BF-BOF 1.370 · DRI-EAF 0.481 · scrap-EAF 0.072

Alloy chain — the stainless differentiator

Input	Factor	Note
HC ferrochrome, global average	2.3 tCO ₂ /t	closed SAF, cleaner power
HC ferrochrome, Odisha captive-coal	5.4–6.2 tCO₂/t	open/semi-closed SAF; captive subcritical coal ≈1.08 t/MWh; ~70–75% of the total is power
Nickel pig iron (contained Ni)	60–85 tCO ₂ /t	Indonesian/Chinese RKEF routes
Class-1 nickel	~13 tCO ₂ /t	
Ferromolybdenum	6.5 tCO ₂ /t ±30%	least certain factor; flagged

India power and fuels (regulator-grade)

Source	Factor	Basis
Grid electricity (FY24-25)	0.710 tCO ₂ /MWh	CEA CO ₂ Baseline Database v21.0
Captive subcritical coal power	1.045 tCO ₂ /MWh	CEA default heat rate 2,500 kcal/kWh, 8% aux
Domestic coal (G12 blend)	1.46 tCO ₂ /t · 92.5 g/MJ GCV · oxidation 0.98	BEE PAT / CEA App. B
LSHS / furnace oil	3.16 / 3.04 tCO ₂ /t (71.9 g/MJ GCV)	BEE / CEA
Natural gas	1.82 kg/SCM (49.4 g/MJ GCV) → –31% vs FO in reheating	CEA App. B
Bio-LDO / green hydrogen	≈ 0 fossil Scope 1	IPCC biogenic accounting

4. Methodology alignment: worldstainless / ISSF

File 06 · ISO 14040/14044, ISO 14404, GHG Protocol, worldsteel LCI

- Standard boundary: **cradle-to-gate** (upstream alloys = Scope 3; melting/refining and downstream = Scope 1+2). Upstream Scope 3 is 60–75%+ of the footprint on virgin/NPI routes — the alloy mix moves total carbon far more than furnace efficiency.
- Recycled-content method** (cut-off): scrap enters burden-free; virgin alloys carry full upstream burden. Used for factory-gate reporting and CBAM. **End-of-life method** (IISI Appendix 5): $X = X_1 + (X_2 - X_1) \cdot RR \cdot Y$ with $RR \approx 0.95$, $Y \approx 0.909$ — reported as our second output.

- Benchmark curve (worldstainless): 85% scrap → 1.95 · 75% → 2.45 · 50% → 3.70 · 30% → 6.80 tCO₂/t. Rule of thumb: each +10pp scrap ≈ −0.8 to −0.9 tCO₂/t.

5. Jindal Stainless: disclosed emissions FY22–FY26

Files 03, 16 · JSL ESG Factsheets, BRSR, CDP, Integrated Annual Reports

Metric (tCO ₂ e/tcs)	FY22	FY23	FY24	FY25	FY26
Scope 1+2 intensity	1.98	2.08	2.15	1.85	1.76
Scope 3 intensity	n/r	1.76	1.90	1.64	1.27
Cradle-to-gate total	–	–	4.05	3.49	3.03

- Absolutes (BRSR): FY24 Scope 1 = 2,992,334 t; Scope 2 (market) = 787,141 t → 3,779,475 t. FY25: 2,995,799 + 622,511 = 3,618,310 t (−4.26%). FY26 Scope 2 market-based 474,893 t.
- Scope 1 ≈ 79% of S1+2 (CDP FY24: 1.70 vs 0.45) — fuel substitution outweighs grid greening alone.
- Levers on record: 70.12% scrap (BRSR FY26); 47% renewables FY26 (315.6 MW hybrid contracted); green H₂ 90 Nm³/h at Hisar, ~600 Nm³/h planned at Jajpur; Bio-LDO at 30% of liquid fuel in Hisar hot rolling (−17,400 tCO₂e/yr); ₹700 Cr solar programme; FY35 target 0.99 (−50% vs FY22); Net Zero 2050.

6. CBAM: regulation, defaults, and the verification dividend

File 08 (§1–7 canonical) · Regulation (EU) 2023/956; Implementing Regulations (EU) 2025/2620, 2025/2621, 2026/1740

Year	Phase-in	ETS €/tCO ₂	India default tCO ₂ /t (with markup)	Default €/t	Jindal verified €/t (1.5 basis)
2026	2.5%	80	7.14	14	3
2027	5.0%	89	7.79	35	7
2028	10%	95	8.44	80	14
2029	22.5%	100	8.44	190	34
2030	48.5%	109	8.44	446	74 (48 on 0.9 direct basis)

The verification dividend. €372–398 saved per tonne by 2030 by filing verified plant data instead of accepting EU defaults. On Jindal's 60–80 kt/yr EU export book: **€22–32M ≈ ₹200–290 crore per year**. The 2029→2030 phase-in jump (22.5%→48.5%) is the financial cliff that makes tooling urgent now.

Notes: for iron and steel the definitive phase prices **direct** emissions only (indirect/electricity excluded while EU producers receive indirect-cost compensation); post-consumer scrap carries zero embedded emissions, pre-consumer scrap carries its origin burden — mass-balance traceability is required to avoid punitive defaults.

7. EU market access beyond CBAM: CPR, ESPR, Digital Product Passport

Files 12, 20 · Regulation (EU) 2024/3110 (CPR), 2024/1781 (ESPR), CEN/CENELEC EN 18216–18223

- **CPR:** from 8 Jan 2026, steel placed on the EU construction market must declare GWP (EN 15804+A2, product-specific, facility-specific) as a condition of CE marking. Generic industry averages no longer permitted.
- **ESPR / DPP:** iron and steel are first-wave priority products; delegated act expected Q4 2026, mandatory compliance ≈2028. The Digital Product Passport is a decentralised, machine-readable record (heat number, chemistry, recycled content, GWP) using EN 18219 identifiers, QR data carriers, JSON-LD/REST interfaces.
- **Green procurement:** the proposed Industrial Accelerator Act would require ≥25% low-carbon steel in EU public procurement from 2029 — no geographic restriction, an open opportunity for verified low-carbon exporters.
- Implication: the calculator's per-tonne record is designed as the natural feeder for EPD/DoPC and DPP data fields — compliance infrastructure, not just an estimator.

8. Jindal's EU exposure: exports, quotas, competitive position

File 19 · CARE Ratings, JSL results, Iberjindal disclosures, UN Comtrade, Goldman Sachs / ICRA / CRISIL

- Sales volume 2.17 → 2.56 Mt (FY24→FY26); export mix deliberately moderated 17% → 8%; estimated direct EU shipments **60–80 kt/yr**, ~30 kt processed through Iberjindal S.L. (Lucena, Spain; ~€60M turnover).
- Mid-2026 EU safeguard revision: tariff-free quota cut 47%; out-of-quota duty doubled to 50%; India-exclusive quota (Cat. 14 bars/sections) 92,557 t. CRISIL: 25–30% decline in Indian steel exports to EU; EEPC: ~\$1.36B revenue at risk.
- Analyst context: Goldman Sachs — Indian BF-BOF steel faces \$102–190/t CBAM cost (15–28% margin compression); ICRA — \$50–140/t through 2034. Jindal's scrap-EAF route with verified data is the structural exception — positioned to fill the supply vacuum as carbon-intensive routes exit the EU market.

9. Competitor tools: Primetals and SSAB teardown

File 07 · product documentation and UX walkthrough

	Primetals CO ₂ Calculator	SSAB EcoUpgraded	StainlessCarbon
Focus	carbon-steel route engineering (BF-BOF/DRI/EAF)	downstream vehicle use-phase	stainless EAF-AOD production
Alloy chain (FeCr/NPI)	not modelled	not modelled	modelled explicitly
Grid context	European defaults	n/a	India grid + captive coal
Feasibility limits	none (100% scrap allowed)	n/a	80% cap + warnings
CBAM €/t	partial (ETS/CBAM cost indicators)	none	full 2026–30 escalator
Access	enterprise-gated; 30+ metallurgical inputs	free app; 5–7 inputs	free web; 4 levers

Positioning: "SSAB simplicity × Primetals depth", rebuilt for stainless metallurgy and India. Five documented gaps: alloy blindspot, all-or-nothing usability, India-context blindness, no practicality constraints, no link to any company's real disclosed trajectory.

10. The scrap ceiling: tramp elements and sensor sorting

File 11 + file 01 §E

- Cu and Sn are noble to iron at 1600 °C — they cannot be oxidised out in the AOD and accumulate linearly with scrap ratio, causing surface hot-shortness (liquid Cu-rich films penetrating grain boundaries during hot rolling; Sn depresses the solidus below 900 °C and worsens it).
- Supply side: India's scrap share is only 23–25% of steel input; 48 countries had scrap export restrictions (Mar 2025); the global scrap market is projected to swing to a ~15 Mt deficit by 2030. Hence the **80% hard cap** in the tool.
- The escape path is at the scrap yard, not the furnace: AI-driven XRF/XRT/LIBS sensor sorting (~20 t/h lines) upgrades tramp-laden obsolete scrap; abatement cost ≈\$30–43/tCO₂ — well below the 2030 CBAM price of €109.

11. Advanced levers (roadmap depth)

Files 09, 10, 13, 14, 15

- **Biochar in ferrochrome SAFs (Odisha):** bamboo biochar (2–5% ash, 81–85% fixed C) is a near-drop-in for met coke; resistivity converges with fossil carbon above ~1400 °C; paddy-straw biochar capped ~10–20% by slag chemistry. 30% substitution abates 0.45–0.66 tCO₂/t FeCr; marginal abatement cost ≈\$29–45/tCO₂. Aligned with the Ministry of Steel's 2024 "Greening the Steel Sector" roadmap.
- **Slag valorisation:** AOD slag self-pulverises ($\beta \rightarrow \gamma$ dicalcium-silicate "dusting") — ideal for mineral carbonation without grinding; processed slags substitute cement clinker (Jindal–Harsco \$150M contract expansion already on record). Dual benefit: Scope 1 mineralisation + Scope 3 avoided emissions.
- **Logistics:** shifting inbound bulk from road (55–101 gCO₂e/t-km) to the electrified Dedicated Freight Corridor (11.5 g) saves ~40–67 kg CO₂e/tcs depending on route basis — scalable at near-zero marginal cost.
- **Internal carbon shadow pricing:** a dynamic CBAM-aligned shadow price inside melt-shop charge optimisation (MILP) flips NPI-vs-scrap procurement decisions before the 2030 cliff arrives.
- **Captive power value stacking:** replacing captive coal with hybrid renewables stacks LCOE savings + CCTS credits + CBAM avoidance + RCO compliance to ≈₹6.6/kWh of combined value — the transition pays for itself.

12. Assumptions, limitations and source register

- Intensities per tonne of **crude** steel; Scope 2 market-based; CBAM module prices direct emissions only (conservative — captive-electricity emissions count as indirect under CBAM and would lower Jindal's basis further).
- Model constants are calibrated so disclosed FY24/FY26 states reproduce audited intensities ($\pm 2\%$); ferromolybdenum $\pm 30\%$; NPI 60–85 t/t range reflects sourcing variation; ETS path is analyst consensus, not a forecast guarantee.
- The 80% scrap cap is a metallurgical-plus-supply judgment for critical grades; sensor sorting can raise it over time.

Primary sources

Jindal Stainless ESG Factsheets FY24–25 & FY26; BRSR FY24–26; CDP Climate Change disclosures; Integrated Annual Reports · CEA CO₂ Baseline Database v21.0 (Dec 2025) · BEE PAT normalization documents and energy-auditor guides · IPCC 2006 Guidelines Vol. 2 (2019 refinements) · MoEFCC BUR-3 · worldstainless/ISSF publications and Fujii et al. (2005) LCI · worldsteel LCI methodology · ICDA ferrochrome LCA · Metso Outotec technical papers · Regulation (EU) 2023/956; Implementing Regulations (EU) 2025/2620, 2025/2621, 2026/1740; Regulation (EU) 2024/3110 (CPR); Regulation (EU) 2024/1781 (ESPR); CEN/CENELEC EN 18216–18223 · Ministry of Steel, "Greening the Steel Sector in India" (2024) · Smart Freight Centre GLEC v3.01 / ISO 14083; DFCCIL operational data · Goldman Sachs, ICRA, CRISIL, GTRI analyses · CARE Ratings; UN Comtrade/WITS · TOMRA sensor-sorting specifications · Orbix / Carbon8 carbonation case studies.

Full research files with inline citations: github.com/ritehrks/jindal-steel-case-study → /research (19 files; research/22-canonical-model-v1.md governs all figures).